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Mission-Level Design Studies for Efficient Hybrid-Electric Regional Aircraft Concepts

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ABSTRACT

This work delves into the design and operation of a series/parallel partial hybrid-electric architecture for regional aircraft. Employing a comprehensive approach, this study leverages mission-level analysis to optimize a 19-passenger hybrid-electric aircraft. The conceptual design framework employed is based on the OpenConcept library, and a systematic computational scheme is developed to effectively investigate the concept's performance, utilizing the supplied and shaft power ratios. Through the examination of three distinct mission ranges and consideration of two technological scenarios, this work offers valuable insights. For the longest mission, an aircraft design optimization problem is posed, and a 23% reduction in total energy consumption is achieved for the optimistic technological scenario. On the other hand, the focus shifts to optimize the power management for shorter missions, where a 26% and a 32% reduction in energy consumption are achieved for the typical and short missions. The results highlight the potential of hybrid-electric propulsion for regional aircraft.

INTRODUCTION

Aviation's environmental impact

The rapid increase of industrialization has led to significant environmental degradation. The impact of human activities on the environment has emerged as a pressing global concern, initiating discussions among the public and governments.

The aviation sector stands out as an important player among the industries contributing to environmental degradation. This sector has long been a cornerstone of the global economy, connecting people and goods across the globe. Covid-19 brought challenges to the aviation industry due to the drop in passengers and routes [1]. However, the world is gradually recovering, and airlines are resuming their operations. This is raising concerns about the industry's contribution to climate change and noise pollution [2-4].

The aviation sector has recognized the urgency of reducing its environmental footprint. Various organizations have set strict environmental goals [5-7]. Achieving these ambitious goals is considered a complicated task since it requires efforts from multiple fronts including research, innovation, and policymakers.

Efforts to enhance the energy efficiency of current aviation technologies have yielded promising results over the past decade. Research efforts have focused on improving existing propulsion and airframe systems, leading to more fuel-efficient aircraft. However, achieving substantial reductions in aviation's environmental impact will require the adoption of disruptive technologies.

One of the most promising solutions is the electrification of aviation. By transitioning from traditional engines to electrified powertrains, aircraft could potentially reduce fuel consumption, emissions, and noise levels, marking a substantial leap towards the environmental goals [8]. However, the challenges of implementing all-electric aircraft across the full spectrum of commercial missions have proven hard to overcome [9,10]. The limitation of current energy storage components, particularly the energy density for long-range missions is one of the main challenges.

Therefore, the aviation sector has turned its attention to hybrid-electric aircraft concepts. These can act as a feasible steppingstone technology, combining electrical energy with fuel to achieve more efficient flight operations. The introduction of electrification concepts opens new possibilities for aircraft designers, offering greater potential to explore innovative airframe and propulsion system designs [11]. Noteworthy technological advancements include the investigation of distributed propulsion systems combined with novel technologies such as boundary-layer ingestion [12].

Moreover, electrification technologies are investigated in a variety of commercial emissions beyond long-haul flights. The short-haul sector holds immense potential for early adoption of disruptive hybrid-electric propulsion technologies. Within this sector, commuter or regional aircraft operate for distances up to 600-800 nautical miles. According to Schäfer *et. al.* [13] electrified short-haul aircraft has the potential to operate for up to approximately 15% of the world's revenue passenger kilometers. The introduction of electrified concepts can revolutionize the regional travel landscape paving the way for further sustainable aviation practices.

However, the transition to hybrid-electric propulsion requires a shift in traditional aircraft design approaches. The complex and interdisciplinary nature of hybrid-electric systems requires a holistic approach that considers various aircraft disciplines and their interdependencies. Detailed mission-level studies can enable engineers to effectively analyze the performance of hybrid-electric aircraft under real-world scenarios, enabling the exploration of design trade-offs to get critical insights into energy consumption and emissions.

The Scandinavian paradigm

The potential use of electrified aircraft in Scandinavia has been gaining attention over the last few years. This market includes several relatively short routes, some of which rank among the most

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trafficked routes in Europe. The OAG Aviation Analytics company published a report [14] with the busiest flight routes globally. The Scandinavian international routes included in the list are summarized in Table 1.

Table. 1 Europe's busiest international routes [14].

Route	Rank	Flight distance	
		km	NM
ARN – OSL	2	385	208
CPH – OSL	3	518	280
ARN – CPH	4	548	296

Furthermore, a report from Nordic Innovation [15] focusing on aviation business models confirms that the introduction of electrified regional aircraft could improve mobility between the countries in this region as well as regional connectivity. Currently, according to [15], only 95 routes serve inter-Nordic connections, meaning that passengers need to travel to the big city hubs to reach their destinations.

An abundance of potential routes within the achievable ranges for electrified aircraft is demonstrated through a point-to-point feasibility study. The airport data utilized in this study are retrieved from the open-source database OpenAIP [16]. To ensure the reliability and relevance of the data, a filtering process is implemented, retaining only operable airports for further analysis. The final count of operable airports within Scandinavia is 88.

The airport latitudinal and longitudinal coordinates are extracted from the database and the Haversine formula [17] is employed to calculate distances between the airports. A histogram of airport distances in Scandinavia is presented in Figure 1.

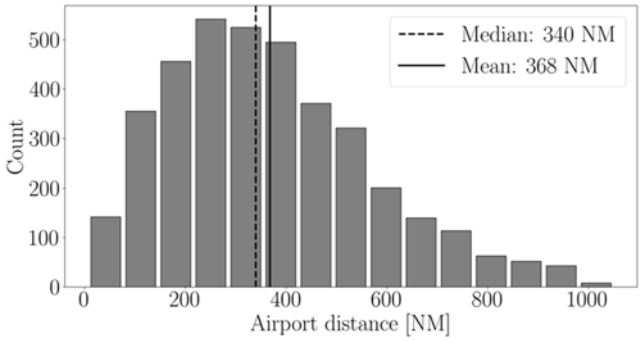


Fig. 1 Histogram of the distance between airports in Scandinavia.

A set of different range test cases are selected. The percentage of covered airports in the whole region as well as within the individual countries are presented in Table 2.

Table. 2 Percentage of covered airports for different ranges.

km	NM	Scandinavia	Sweden	Denmark	Norway
370	200	21.9%	37.5%	90.5%	42.8%
833	450	69.2%	84.7%	100%	80.1%
1111	600	86.8%	93.3%	100%	88.4%

Furthermore, a visual representation of the Scandinavian map and the operable airports is shown in Figure 2. The marker color corresponds to the number of reachable airports from each airport for a mission of 450 NM.

The data confirms that a significant portion of the airport routes in Scandinavia can be adequately serviced by regional aircraft. It is worth noting that certain routes, particularly those in the northern region of Norway would require aircraft capable of flying at high altitudes to prevent interference with the landscape. However, this consideration falls outside the scope of this work as it primarily focuses on providing a preliminary overview of the potential of electrified aviation within the region.

Finally, all Scandinavian countries are committed to adopting more sustainable practices and mitigating environmental impact. Their comprehensive strategies involve substantial reductions in national Greenhouse Gas emissions by 2030 and a long-term goal of achieving climate neutrality by 2050 [18]. Given the established electric transport infrastructure and network of small airports, Scandinavia presents an ideal location for the commercialization of electric aviation.

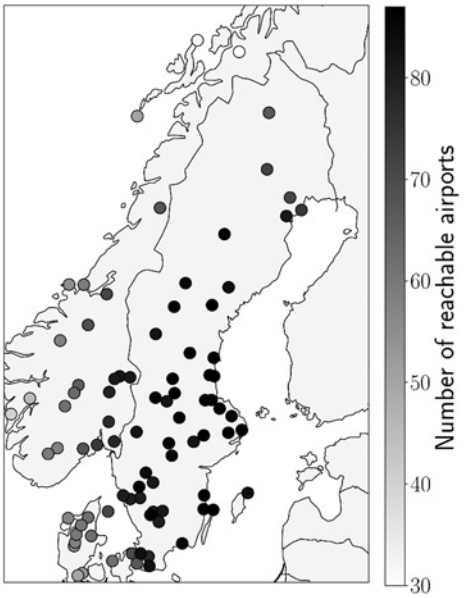


Fig. 2 Geographical distribution of reachable airports within 450 NM (833 km) of operable airports in Scandinavia.

Present work

Driven by the opportunities within in the Scandinavian region, where regional aircraft can efficiently serve a significant portion of airport routes, the present work aims to investigate a hybrid-electric 19-passenger aircraft. The analysis covers flight ranges of 200 NM, 450 NM, and a design mission of 600 NM, aligning with the Flightpath 2050 goal of enabling 90% of European travelers to complete journeys within four hours [5]. Using a two-stage approach involving low-fidelity models and mission analysis, the study allows for both rapid concept evaluation and a thorough assessment of the operational viability. A noteworthy innovation is the integration of various mission studies and distinct optimization strategies, an approach that emphasizes the essential importance of considering operational factors throughout the investigation. This approach not only uncovers design details but also emphasizes the importance of operational factors, paving the way for further progress in future research.

METHODS

This work employs the OpenConcept [19] library to model a hybrid-electric 19-passenger aircraft following the CS-23 certification [20]. OpenConcept is a conceptual aircraft design tool, which includes different aircraft discipline calculations and integrated into a design framework. The library is built on top of the optimization library OpenMDAO [21], enabling the design space exploration and optimization for different aircraft concepts including electrified applications.

The framework consists of low-fidelity models for aircraft components' performance and weight estimation, including airframe, engines, propellers, etc. The aircraft weight estimation model is based on empirical correlations from Roskam [22] and Raymer [23]. The library utilizes a non-linear Newton solver to compute the state variables and attain zero residual values. The user can determine these variables by developing the appropriate computational scheme.

Finally, it is important to note that each component model

implemented contains efficient gradients to enable quick and accurate multidisciplinary design optimization.

Hybrid-Electric Computational Scheme

The aircraft under investigation utilizes the series/parallel partial hybrid electric architecture. The propulsion system is comprised of two conventional turboprop engines and two electrically driven propellers (e-propellers) – one per wing. In addition, a battery energy storage system is available. The e-propellers can be driven either by battery or supplementary power from the traditional engines (denoted as power off-take). This is ensured by introducing a Power Management and Distribution (PMAD) system. The main architecture components are shown in Figure 3.

This architecture is considered as complex mainly due to the presence of two energy sources overall and two possible thrust generation locations per wing. Therefore, the development of a consistent computational scheme is of utmost importance. The scheme used in this work is built upon the framework developed by the authors in [24]. However, the scheme is further developed for the purposes of this work. The enhancement of the controlling parameters tailored specifically for this architecture is presented here. The main driver of this development is to significantly improve the understanding and practical implementation of this architecture.

Within the present scheme, the parameters controlling the scheme calculations are the powertrain components' throttle, the supplied power ratio, and the shaft power ratio. The definitions are based on [25, 26]. The definition of the supplied power ratio refers to PMAD component and presented in Eq. (1).

$$\Phi = P_{bat}/(P_{bat} + P_{off,1} + P_{off,2}) \quad (1)$$

where P_{bat} is power provided by the battery and P_{off} is the power delivered to the generators from engines 1 and 2. Note that for this definition, $\Phi < 0$ denotes that the battery is charging. However, battery charging is not considered in this study.

Regarding the shaft power ratio, the power levels in Eq. (2) are not the useful propulsive power but the power delivered from the shaft to the propulsor.

$$\phi = P_{es}/(P_{es} + P_s) \quad (2)$$

In Eq. (2), P_{es} is the power delivered to the e-propellers by the motors, and P_s is the power delivered to the propellers by the engines. Both ratios presented in Eq. 1 and 2 are predetermined and can vary at different levels throughout the mission but remain constant within each mission phase.

The implicit gaps depicted in Figure 3 are incorporated to enable the solver to adjust the throttle levels of the components. The engine and motor component throttles are presented in Eq. (3). The throttles are defined as the ratio between the power delivered by each component divided by their rated (maximum) power.

$$\xi_{GT} = P_{GT}/P^*_{GT},\ \xi_{EM} = P_{EM}/P^*_{EM} \quad (3)$$

The throttle level of the motors is determined to maintain stable flight conditions at each intermediate point of the mission. At the implicit gap between the generator and the PMAD, the engine power delivered to the generator P_g is selected to meet the power requirements of the motors for a predetermined Φ. Simultaneously, the engine throttle is adjusted to fulfill the propeller power P_s requirement calculated based on the ϕ.

A mission profile including a main mission as well as a diversion mission of 185 km (100 NM) based on the certification requirements is constructed by linking the flight segments. In addition, a loiter phase including 30 mins of steady flight at 457 m (1500 ft) using only one energy source is implemented according to the CS-23. The equations of motion of the aircraft are integrated into the mission analysis.

The mission analysis is based on a numerical quasi-steady point model where each mission phase consists of a series of control points in space and time. For every control point, the horizontal and vertical speeds are predefined to satisfy the aircraft design assumptions. Therefore, the library calculates the lift coefficient and propulsors' throttle to balance the lift and drag forces. High-level parameters of interest such as fuel burn and energy consumption are calculated using numerical integrators over the mission control points and time. The aircraft weight is updated at each control point based on the fuel burn integration calculation.

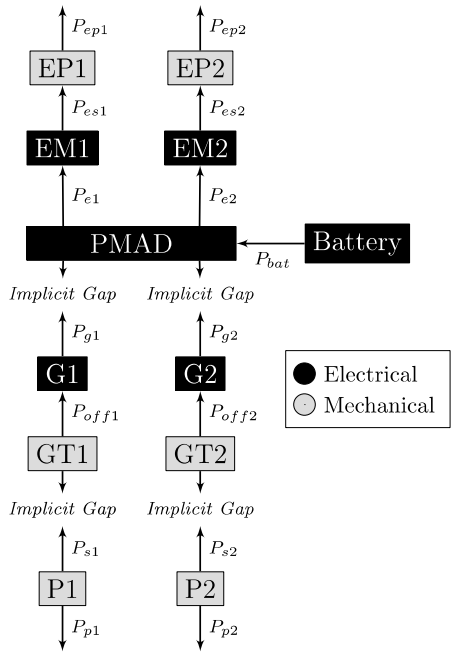


Fig. 3 Series/Parallel Partıal Hybrid-Electric architecture.

Technology Overview

Firstly, a conventional 19-passenger aircraft model is constructed within the OpenConcept library. The conventional aircraft model utilized data from the Beechcraft 1900D aircraft [27]. Using this model as a baseline, a 19-passenger hybrid-electric model is developed and investigated.

The hybrid-electric aircraft is examined for an Entry-Into-Service (EIS) of 2035+. These are reflected by the aircraft model assumptions summarized in Table 3. The engine performance is based on the Pratt & Whitney PT6A-67D unit [28]. The Maximum Take-off Weight (MTOW) upper limit is set at 8618 kg based on CS-23.

Table. 3 Aircraft model assumptions and requirements.

Parameter	Value
PSFC (cruise)	0.094 g/(kW*s)
MTOW	≤8618 kg
Passengers	19
Payload (Weight incl. baggage)	19*99 kg (87kg + 12kg)
Cruise altitude	3048 m (10,000 ft)
Loiter altitude	457 m (1500 ft)
Cruise Mach speed	0.35
Rate of Climb (MTOW, SL, ISA)	≥ 6.35 m/s
Approach speed	≤ 62 m/s

The aircraft system-level estimations are sensitive to technology assumptions. Therefore, two scenarios are defined for this work.

The electrical components are considered both through their efficiencies within the chain of components as well as their specific power. The PMAD is a simplified electric system where electrical components such as inverters, converters and rectifiers can be used depending on the power level and type of current.

Table 4 summarizes the electrical component assumptions for both technology scenarios. The assumptions are based on the work of Nasoulis *et. al.* [29] and Zamboni *et. al.* [30] and are confirmed by the technology forecast presented in [31, 32] considering novel lithium battery chemistries.

Table. 4 Technology scenarios.

Parameter	Unit	Scenario #1	Scenario #2
Battery SE (Cell)	Wh/kg	700	1000
Battery SE (Pack)	Wh/kg	395	565
Battery SP	kW/kg	0.7	0.9
Motor SP	kW/kg	10	15
Generator SP	kW/kg	12	20
PMAD SP	kW/kg	19	25
Motor Eff.	-	0.95	0.98
Generator Eff.	-	0.90	0.95
Battery Eff.	-	0.85	0.90
PMAD Eff.	-	0.90	0.95

Optimization Framework

The hybrid-electric aircraft presented in the previous subsections is optimized for a 600 NM mission. The optimization problem for aircraft design is summarized in Table 5. The optimization objective is the total energy consumption during the main mission (block energy). This includes both the electrical energy and the chemical energy from the fuel. To calculate the chemical energy, the estimated fuel burn is multiplied with the assumed fuel specific energy (43 MJ/kg).

Table. 5 Design optimization problem for hybrid-electric aircraft.

	Variable	Quantity
Objective	Block Energy	1
Design Variables	MTOW	1
	S_{ref}	1
	W_{batt}	1
	N_P	1
	N_{EP}	1
	D_P	1
	D_{EP}	1
	P^*_{GT} (rated)	1
	P^*_{EM} (rated)	1
	P^*_G (rated)	1
	Φ	6
	ϕ	7
Constraints	$0 \leq R_{MTOW} \leq 5$	1
	$0.2 \leq SOC_{loiter}$	1
	$0.1 \leq \xi_{EM} \leq 1$	7
	$0.2 \leq \xi_{GT} \leq 1$	7
	W/S = 345 kg/m^2	1
	V_{tip} = 200 m/s	2
	P/D^2= 200 kW/m^2	2

The MTOW, wing area (S_{ref}), battery weight (W_{batt}), powertrain component rated powers, supplied and shaft power ratios (Φ and ϕ) are varied by the optimizer to minimize the total energy consumed. The rated engine and electrical component powers are varied based on the maximum power delivered by each of them throughout the mission. Finally, Φ is selected to be varied only during the main and diversion mission. During the loiter phase, the aircraft operates exclusively on a single energy source, and therefore in turboelectric mode as defined in [33]. Consequently, the Φ factor is set to zero and as detailed in Table 5, there are a total of 7 phases in the mission, with Φ considered only for 6. The decision to adopt this configuration is grounded in the guidelines provided by CS-23. Regarding the propellers' design, the propeller and e-propeller diameter are selected by the optimizer for loading of 200 kW/m^2 based on estimations from [34]. Both propeller and e-propeller are fixed-speed, and their rotational speeds are calculated in order to maintain a maximum propeller tip speed of 200 m/s.

It is important to highlight that the MTOW residual (R_{MTOW}) is not set to zero by the solver but is rather exposed to the optimizer as an inequality constraint. This residual compares the design variable MTOW against the calculated Operating Empty Weight (OEW), total fuel weight, battery weight, and payload. In addition, a lower limit for the battery state of charge at the end of the mission SOC_{loiter} is set to 20% to ensure safe battery operations. For the rest of the powertrain components, the motor and engine throttles are kept within feasible levels. With the engines' and motors' rated power serving as design variables, and the throttles upper limit set to 1, a sufficient component sizing for meeting maximum power requirements throughout the mission is ensured. Finally, the implementation of state-of-the-art materials and advanced manufacturing processes for EIS is anticipated to result in higher wing loading thresholds. The improved aircraft technology level is considered during the optimization process by keeping the wing loading (W/S) at constant levels (345 kg/m^2) compared to the conventional aircraft (270 kg/m^2).

The integrated aircraft design framework is presented in Figure 4. The framework follows the workflow presented and discussed in [35]. The optimizer sets the design variables (x) presented in Table 5 and passes them on to the aircraft model. For example, the Φ and ϕ factors are determined by the optimizer for different mission phases. At the same time, a set of fixed design parameters is set by the user and passed to the aircraft model. These parameters include both the top-level aircraft requirements presented in Table 3 as well as the scenario assumptions summarized in Table 4.

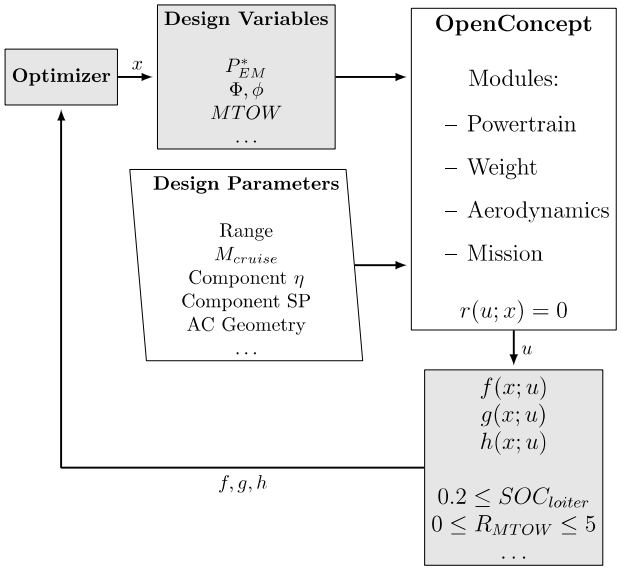


Fig. 4 Integrated aircraft design framework.

The model simulates the given aircraft and is comprised of different modules as shown in Figure 4. It also contains a non-linear Newton solver. The $r(u;x)=0$ denotes that the x is fixed by the optimizer and u is iterated by the Newton solver to reduce the residuals towards zero. When the model converges, u state variables become fixed, and the objective function (f) and constraint functions – inequality (g) and equality (h) – are calculated. Their calculated values return to the optimizer as shown in Figure 4. The $f(x;u)$ expression denotes that the model is converged and fixed

while the optimizer searches for the next value of x.

When the aircraft design is finalized for the 600 NM range certain aircraft parameters are established. However, the aircraft is also capable of operating within shorter ranges, specifically the typical range of 833 km (450 NM). In such instances, an additional optimization problem arises to determine the optimal operational approach. The summary of the operational optimization problem can be found in Table 6.

The power management strategy is adjusted throughout the mission to minimize the total energy consumption during the main mission. In order to maintain safe operations, the state of charge (SOC) and component throttles are limited to appropriate levels as before. Finally, it is expected that the Take-off Weight (TOW) will be lower for shorter missions. However, it is not possible to know the TOW *a priori*. The TOW selection affects the aerodynamic performance and hence the fuel consumption. Therefore, the TOW is selected as a design variable and its residual compares the TOW value against the OEW, total fuel, payload, and battery weight. The TOW residual is kept in low levels throughout the optimization.

Table. 6 Power management optimization problem for hybrid-electric aircraft

	Variable	Quantity
Objective	Block energy	1
Design Variables	Φ	6
	ϕ	7
	TOW	1
Constraints	$0 \leq R_{TOW} \leq 5$	1
	$0.2 \leq SOC_{bat,loiter}$	1
	$0.1 \leq \xi_{EM} \leq 1$	7
	$0.2 \leq \xi_{GT} \leq 1$	7

The Interior Point Optimizer (IPOPT) optimizer [36] is selected utilizing the pyOpt library [37] for both optimization problems. The use of a gradient-based optimizer is enabled due to the design philosophy of the OpenConcept as discussed in [19].

The Productivity Specific Energy Consumption (PSEC) metric is employed for the different technology scenarios and mission ranges. The definition of this metric is presented in Eq. (3) and is based on [38]:

$$PSEC = \frac{m_{fuel} * LHV + E_{el}}{Payload * Range} \qquad (3)$$

where m_{fuel} is the total fuel consumption for the main mission and LHV is the fuel lower heating value (assumed 43 MJ/kg) and, finally E_{el} is the electrical energy consumption for the main mission. The payload is constant for all cases under investigation. The PSEC metric has units of kJ/(kg*km).

RESULTS

The methodology described in the previous section was employed for the aircraft under investigation. After performing a series of optimizations, results from the optimal solutions are extracted and presented in this section. In parallel, simulations for conventional Beechcraft 1900D aircraft are performed. The respective hybrid-electric aircraft parameters presented here are relativized against the conventional values.

It is crucial to account for the fidelity of the models and technology assumptions employed within this study. These factors directly impact the accuracy of the results presented here. The primary focus of this work is not to provide precise predictions for individual aircraft designs. Rather, the goal is to compare powertrain architectures within a specific aircraft class, ensuring consistency in modeling assumptions for a fair comparison. Despite the limitations, the trends in electrification remain consistent across the series of missions and technology scenarios examined, indicating the general applicability of the results.

Aircraft Configuration and Design Analysis

Within this subsection, the design analysis of hybrid-electric aircraft is discussed. A comparison of the aircraft design results for different cases is summarized in Table 7. In addition, the weight breakdown for both scenarios and conventional aircraft is presented in Figure 5.

The hybrid-electric aircraft configurations are heavier due to the additional components required to facilitate the use of electrical energy. While the structural weight of the aircraft is moderately lower for the hybrid-electric case due to the technology assumptions discussed above, the addition of the electrical power system (EPS) system and battery results in an increased OEW as presented in Table 7. In addition, it is important to highlight the high contribution of the battery weight in both hybrid-electric aircraft designs as well as the results for the Maximum Landing Weight (MLW). The high share of the battery weight in both cases is of major importance since batteries do not considerably modify their weights as they discharge. This factor affects the aerodynamic performance as the hybrid-electric aircraft weight is changing at a lower rate while flying.

Table. 7 Aircraft design results against the conventional aircraft.

Variable	Scenario #1	Scenario #2
Rel. MTOW	2.3%	2.9%
Rel. OEW (incl. W_{bat})	19.8%	22.6%
Rel. MLW	4.1%	5.9%
$W_{bat}/$MTOW	11.0%	12.7%

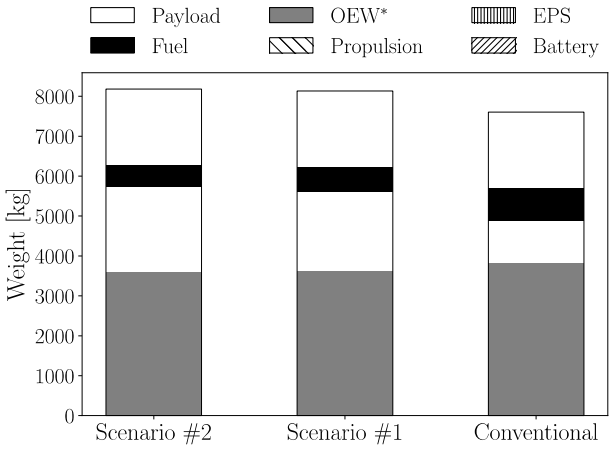


Fig. 5 Aircraft weight breakdown for all aircraft considered. (OEW*: OEW excl. propulsion and EPS)

Operational Characteristics

A series of different mission case studies are summarized in Table 8. The three mission ranges are denoted as short, typical, and long. It is important to highlight the case studies are not based on the same optimization problem. The long mission is optimized using the design optimization problem as presented in Table 5. The aircraft design parameters are then extracted and used for the typical and short missions. For these cases, the optimization includes only operational aspects and the TOW estimation as presented in Table 6.

Table. 8 Case studies description.

Mission Type	Range		Optimization Mode
	km	NM	
Short	370	200	Operation
Typical	833	450	Operation
Long	1111	600	Design

The mission profile under consideration is presented in Figure

6 for the typical mission. The number of intermediate points selected for each phase is 21 and it was selected by performing a sensitivity analysis between the number of points and high-level mission results.

The horizontal and vertical speed for each mission intermediate point is predefined and remains constant through optimization for both technology scenarios. The main mission duration for the typical range is 2.8 hours.

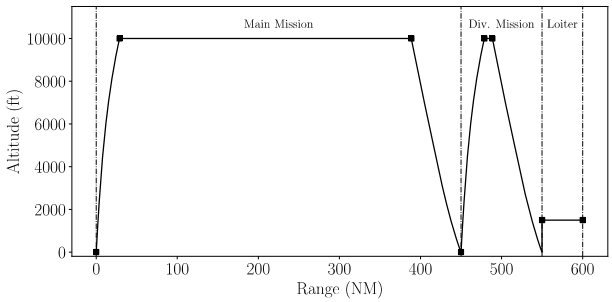


Fig. 6 Typical mission profile.

The high-level parameters of interest for the hybrid-electric aircraft under scenario #1 are presented in Figure 7. The fuel and energy consumption for the main mission are presented as relative values against the conventional aircraft values for the corresponding range.

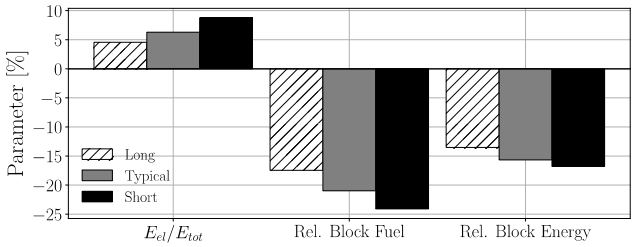


Fig. 7 Results for different missions –Scenario #1.

The introduction of electrical energy results in both lower block fuel and block energy compared to conventional aircraft. However, the block energy reduction is lower compared to the block fuel reduction for all examined missions. This is explained by the introduction of additional powertrain components to facilitate aircraft electrification. A milder block energy reduction is worth noting considering that it refers to both chemical and electrical energy consumed. Within this study, an assumption is made that the battery is fully charged at the start of the mission. However, in reality, the battery is charged either through airport infrastructure or potentially during previous flights. A considerable energy reduction during the flight is an important benefit if considering a "well-to-wing" approach. However, this aspect is currently out of the scope of this study.

An interesting trend observed in Figure 7 is that the short mission presents the largest reduction in terms of fuel and energy consumption. However, the aircraft design and, hence, battery size are optimized for the long mission. This can be explained from the available battery on-board and by observing the ratio between electrical energy and block energy consumption which denotes the degree of hybridization (DoH). The DoH is lower for the long mission and therefore a milder reduction of block fuel and block total energy consumption is achieved. It is important to note that for scenario #1, the SOC at the end of the mission for all examined cases was close to the lower limit set by the optimization (20%). Therefore, it is safe to conclude that the size of the battery is sufficient for the energy requirements of the short mission.

The results for scenario #2 are presented in Figure 8. Again, it is shown that scenario #2 results in lower block fuel and block energy both compared to the conventional aircraft and scenario #1. This is expected due to the combined effect of improved battery SE as well as the electrical component efficiencies leading to higher main mission DoH for all case studies. Similar trends are observed for this scenario as well. It is important to note that for this scenario, the SOC at the end of the mission reaches its lower limit only for the long and typical cases.

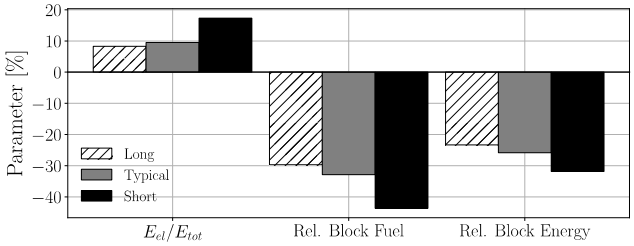


Fig. 8 Results for different missions – Scenario #2.

The PSEC is an indicator of how efficiently an aircraft can transfer passengers from point A to point B with a lower PSEC indicating a more efficient operation. According to Figure 9, scenario #2 is more efficient due to lower block energy consumption for all missions. The trends between different missions observed in Figures 7 and 8 are also confirmed here. The scenario #1 is the least efficient during the long mission as discussed before.

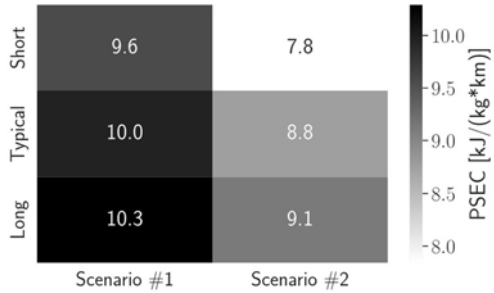


Fig. 9 PSEC metric for different case studies.

The shaft power ratio ϕ for scenario #1 is shown in Figure 10. When examining the shaft power ratio ϕ, it is also important to consider the total power requirements for each mission phase. Therefore, the power breakdown for the long mission is presented in Figure 11. Note that for the rest missions, the power requirements are moderately lower compared to the long mission because of the lower TOW and, hence, aircraft drag.

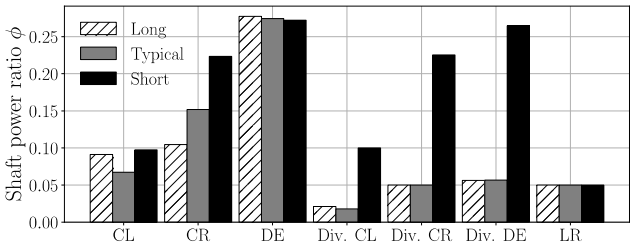


Fig. 10 Shaft power ratio ϕ – Scenario #1.

The trends presented in Figures 10 and 11 are affected by two major factors: the hybridization selected by the optimizer and the imposed limits for components' throttle during the optimization. The climb and cruise phases are considered the most impactful for all missions. For the climb, this is because of the high aircraft power requirements while for the cruise it is because of its long duration. The energy levels from different sources and for each mission phase are illustrated in Figure 12. The percentages shown in Figure 12 are the DoH for each mission phase. Since the optimization objective

is the energy consumption during the main mission, it is crucial to examine the phase durations. For the long mission, the climb lasts for 6.8min, while the cruise lasts approximately 138min based on the model calculations. Therefore, a lower ϕ level for cruise leads to a higher share of energy during the mission and therefore has a higher impact in fuel and energy consumption.

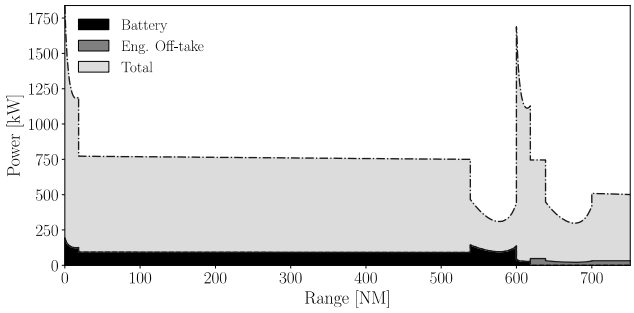


Fig. 11 Power breakdown from different sources – long mission and scenario #1.

Regarding the optimized power management strategy during the descent phase, the utilization of the battery is a consequence of adhering to specific throttle limits set for the engines and motors, which have been designed to avoid any component throttles falling below a threshold as shown in Table 5. During the main mission, the descent phase relies on the battery as its power source. However, during the diversion mission, the e-propellers draw power from the engine and operate at reduced levels, maintaining an optimal balance between conventional and electric propellers.

It is observed that the long and typical mission has similar ϕ, levels while the short mission showcases considerably higher ϕ levels. This trend is in agreement with the block fuel and energy results presented in Figure 7.

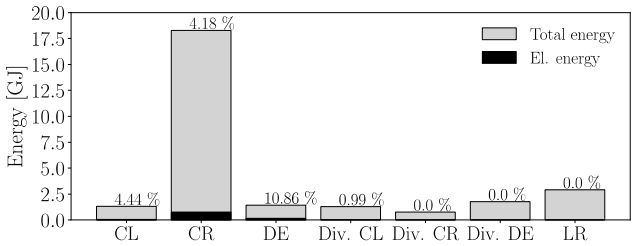


Fig. 12 Energy from different sources and DoH – long mission and scenario #1.

The scenario #2 shaft power ratio ϕ and aircraft power breakdown for the long-range mission are presented in Figures 13 and 14 respectively. The considerable improvement in the battery characteristics is reflected in higher values of ϕ across the main mission. The use of battery is again prioritized for the climb phase due to its high-power requirements and for the cruise phase due to its long duration.

A lower shaft power ratio ϕ during the climb is observed for the typical mission in both scenarios according to Figures 10 and 13. The resulting shaft power ratio is influenced by the selected range and the nature of the optimization problem. Specifically, for the typical and short missions, the optimization relies on a predefined battery capacity optimized for the long mission. When comparing the long and typical missions, the typical mission has a lower Take-off Weight (TOW), resulting in lower climb power requirements. Therefore, a larger portion of the battery's energy capacity is allocated to the cruise phase, which is characterized by its extended duration. On the other hand, when comparing the short and typical missions, the shorter mission range provides the possibility to utilize the battery throughout the entire flight, with no emphasis on preserving energy for extended cruise phases.

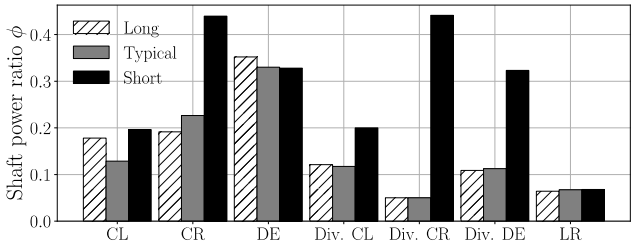


Fig. 13 Shaft power ratio ϕ – scenario #2.

Finally, the lower energy consumption for scenario #2 is confirmed by comparing the energy levels in Figures 12 and 15. The energy consumed during the cruise is considerably lower for scenario #2 and the DoH is approximately double compared to scenario #1. This observation confirms the benefits of electrified aircraft for the case studies examined while highlighting the cruise phase as the most promising in terms of energy reduction.

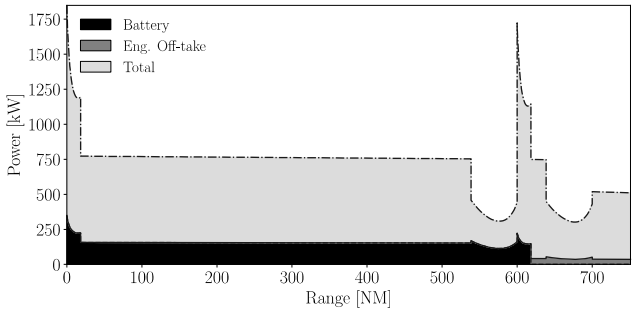


Fig. 14 Power breakdown from different sources – long mission and scenario #2.

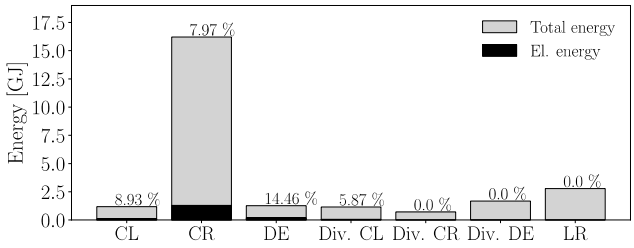


Fig. 15 Energy from different sources and DoH – long mission and scenario #2.

CONCLUSIONS

Within this work, the series/parallel partial hybrid aircraft concept is investigated in the context of regional aircraft operations within Scandinavia. Utilizing a modular framework based on OpenConcept, extensive mission-level analyses were conducted for a 19-passenger hybrid-electric aircraft, under CS-23 certification.

The work's findings underscore a considerable potential for hybrid-electric regional aircraft operation in Scandinavia. Notably, nearly 70% of operational airports in the region can be efficiently accessed within a mission of 833 km (450 NM), highlighting the role of hybrid-electric aircraft in enhancing regional connectivity.

The developed computational scheme provides a systematic approach to minimize block energy consumption. Through a comprehensive investigation of various scenarios and mission ranges, the work offers insights into aircraft performance across varied operational conditions. Developing optimization strategies for different ranges emerges as a pivotal aspect, affecting the observed trends and results. This underscores the necessity of diverse approaches when evaluating hybrid-electric aircraft performance.

While the design optimization achieves a 23% energy reduction

at an 8% hybridization level, exploring shorter missions uncovers further energy savings. Noteworthy is the remarkable 32% decrease in total energy consumption in the second technological scenario for a 200 NM mission, achieved through an average 17% hybridization. This underscores the potential of hybrid-electric propulsion across diverse operational scenarios, reaffirming its role as a transformative force in regional aviation.

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